



OPEN Advanced financial security system using smart contract in private ethereum consortium blockchain with hybrid optimization strategy

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In traditional stock market, the global processing framework shares the data to various mediators like financial and government institutions. The institutional firms needs to handle with large number of data in the system and cooperates with others to provide the trades on the stock exchange platform, and consistently buy-sell orders pass through various parties before settlement. It involves a complex chain of intermediaries, has several drawbacks like weak transparency, long lead times for financial settlements, and a single point of failure. Blockchain (BC) computer node network securely shares the common ledger without intermediaries. This paper uses a deep learning-based Smart contract in the private ethereum consortium blockchain (PEC-BC) to provide financial security. First, the data is collected. Then it is given to the next stage. The Dynamic Butterfly-Billiards Optimization Algorithm (DB-BOA) is used to choose the leader block. Further, the selected new leader block is used in the Adaptive Deep Temporal Context Networks (ADTCN) with a consensus algorithm to make secured smart contracts. Here, the parameters are optimized by DB-BOA. The developed ADTCN-based financial security system was compared with other conventional methods, and algorithms performed well.

Keywords Financial security, Smart contract, Private ethereum consortium blockchain, Dynamic butterfly-billiards optimization algorithm, Adaptive deep temporal context networks

In the future, it is expected to happen the most subsequent shocks and dangerous attacks. The most problematic attacks are integrity corruption of financial data, like transactions, algorithms, and records. Some technical solutions are introduced for such attacks, which have the capacity for confidence and more potential for trust¹. Behind these attacks, malicious actors are included, not just sponsored attackers but bold criminals like the Carbonak group, who target financial institutions for theft². There are various drawbacks and limitations to the conventional stock exchange platforms due to the centralization architecture³. The stock exchange and broker participating system lead to failure with data management inconsistency for each system, resulting in errors and failure and causing more recovery time⁴. The system's limited access affects the data audition and transparent user access, and it takes more time to perform financial settlements⁵. After the trading, it takes almost three days to perform the settlement.

To label the above restrictions, matching engines, and trade executions are built in the financial sectors by the main vendors⁶. The matching technology vendors and one of the leading trading Nasdaq present end-to-end trading processes, managing and hosting product services⁷. In the settlement and trading process, a surveillance system is introduced to regulate and monitor the system for clients. However, there are limitations to this approach, such as a limited level of data transparency, long settlement time, and trading hours⁸. An important thing is the client needs the hardware specification or to follow architecture enforced, which involves multiple organization trading processes in the system distribution to manage the system separately⁹. The system that differs from the Nasdaq provides clearance solutions and settlement for specific cash regulations from some stock markets¹⁰. As a result, multiple systems have been maintained, and the participating system has the chance of getting failure at some point, increasing the overall trading platform complexity in the system architecture¹¹. The blocks are the set of transactions in the linked block tree called a BC. The genesis block is the root of the tree¹².

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Edge computing is the data processing, core function application, network integration, and storage to an open platform that provides the object data source near the closest end-of-page service to meet real-time security, application intelligence, and privacy sexual needs¹³. The Internet plays an important role in developing advanced data processing, and the fundamental technologies support realizing the Internet of Things¹⁴. For the technical purpose, there is a need for computers. The breakthrough of the advanced technology for the Internet of Things can effectively handle the technologies in the local device. The computer system in the local angle carried out the processes¹⁵. It solves the user's individualized needs and provides quick services because it is more user-friendly. The solution for the most critical challenges in BC is continuous innovation in BC development¹⁶. The most used payment protocol is the Lightning Network to overcome the scaling problem in the cryptocurrency and overwrite the existing BC.

The important function of this developed system is presented as follows.

- To create an effective smart contract-based financial security system using the hybrid heuristic-based block-chain technology for making the system helpful in managing the financial or economic crisis.
- To optimally select the leader block in the BC using developed DB-BOA, where further used to reduce computation time, communication cost, and size of the memory.
- To design the optimized algorithm named DB-BOA by the combination of two conventional algorithms as DBOA and BOA for estimating the final solution.
- To develop an ADTCN model for performing the smart contract, where DB-BOA optimally tunes the hyper-parameters like epoch count, hidden neuron, and steps per epoch for attaining maximum accuracy, precision, MCC, NPV and minimizing FPR.
- To validate the better security of the model by comparing the developed DB-BOA with other traditional algorithms and classifiers.

This included a section of the paperwork is given as follows. The existing tasks of the developed model are described in Part II. The financial security system based on BC is explained in Part III. The implementation of PEC-BC is shown in Part IV. The leader block selection using developed DB-BOA is demonstrated in Part V. The ADTCN for security improvement is presented in Part VI. The outcomes and their descriptions are illustrated in Part VII. The final Part VIII summarizes the presented work.

Existing works Related works

In 2020 Abdallah et al.¹⁷ have suggested an architecture of a novel-based BC for the fully decentralized stock market. Also, the architecture was created by the consortium network leveraging organizations acts as validation nodes. While noticing the market regulation in the existing government, the smart contract was implemented by the logical trading of stock exchange mechanism in the architecture. This work encouraged the strategy of decentralized stock market platforms as the new platform did not change the stock market trading logic significantly or remove any parties. Platform's evaluation access by the virtualization-based test network and trade exchange smart contract implemented by the concept proof of new architecture. The virtual nodes of the test network run the developed smart contract stock exchange to measure the different size network latency and network workload. The result showed the sending rate optimal throughput equivalent to the developed trading platform, which reaches the throughput. For the Singapore stock market, the major stock exchange requirement was sufficient by the throughput.

In 2020 Tsoulas et al.¹⁸ have designed a Python decentralized application model for which the BC data was stored in a Neo4j database graph. They developed the Casper-like consensus mechanism based on the principle of the Ethereum BC network, and the efficiency was tested for finality. Proof of stake protocol and proof of workers were employed in the block proposal, and the participant's consensus and incentive criteria differently from each type. The functionality of the BC component incorporated with the graph model was a major part of the work. It was important to analyze the relationship between the true value data extracting for exploiting the benefits of data. This method visualized the BC data changed and monitored in used case scenarios. Here, a series of simulated findings were validated to prevent the BC attack. The most common BC attacks were Catastrophic Crashes, Attacks and dynamic validator attacks. They show the distributed graph in the BC data modelling assists the security enhancement and protocols operation and facilitates to save the information in the path-dependent.

In 2022 Wang et al.¹⁹ have developed a system to increase changes in the development of business models. Such as serving the real economy in the bank as a pioneer, and for the small-sized enterprises, the developed supply chain was used. The bank's development and transformation and the logistic technology promotion of the enterprise's financing were used. To achieve application intelligence, privacy, and security, edge computing form an open platform by data processing, core function application, end-page service, integrated network, and storage. Supply chain financing could be effectively controlled by the optimized plan established by the financing supply chain. The supply chain settlement cost was to be solved by the financial literature integration of the supply chain. BC technology as well as financial supply chain were analyzed by theoretical research. The supply chain financing and BC supply could be combined with specific situations for the cash flow and system management to focus the control system. It reduced the risk of all parties in the financial system. The advanced data processing and the environment based shared data in the BC Iota had practical and theoretical development of enterprises and commercial banks.

In 2022 Amponsah et al.²⁰ have developed a financial security system. National Health Insurance Scheme (NHIS) financial security was threatened by the problems related to data and fraud corruption. If the problems were unattended, the core aim of universal healthcare service would not be realized. To protect the NHIS from

financial plummeting, the BC was implemented based on the design of the work. The paper presented the proposed view of the data management framework, sequence and use case diagram, smart claim processing system, and smart notification system. The successful model of McLean & DeLong's information system detailed the system. The behavioral aspect of the system was validated by using this system. The study discovers that user satisfaction and quality of information in a significant influence.

In 2022 Agyekum et al.²¹ have developed a proxy re-encryption approach for secure data sharing in cloud environments. Encrypted data outsource by the data owners to the cloud by encryption based on identity, while the user access data constructed by the proxy re-encryption. The Internet of Things was an edge device resource-constrained to a proxy server for handling intensive computation. For the effectiveness of the proxy, the feature of information-centric networking was used in the cached content delivery. This improved the use of network bandwidth and quality of service. In data sharing, disruptive technology enables decentralization based on the BC system model. It attained control access to fine-grained data and mitigated the bottleneck in a centralized system. The secure evaluation and the analysis show the approach's security, integrity, and data confidentiality.

In 2022 Shrivastava et al.²² have suggested the Modified Infinite Chaotic Elliptic Cryptography (MICEC) based on the BC to increase cloud security. It contained three phases, which were ownership protection, protection of authentication, and validation of identity mapping. Authentication was the first process for employing the MICEC, where the modified chaotic neutral network and infinite cryptography elliptic curve were combined for data encryption and key generation. The digest 5 algorithm also improved the message generated by the hash function. Then the score value computed by the Latent Dirichlet Allocation with the Recommendation algorithm improved message content. The developed work was evaluated using JAVA programming language with NetBeans IDE. The evaluation measures was simulated and compared with other methods.

In 2023 Li et al.²³ have implemented the secure storage based on BC and designed the Fabric-SCF using distributed consensus to realize data traceability, immutability, and security. The access control model based on attributes was deployed for efficient operation by ensuring the smart contracts defined the system, access control, and system process. Effectiveness designed by the two sets of validation outcomes for verifying the Fabric-SCF performance. As a result, Fabric-SCF attained dynamic and high throughput maintained by the access fine-grained in simulated operation scenarios of the real world.

In 2022 Ahamad et al.²⁴ have proposed a BC technology that develops harmony and trust by communicating directly between two parties without involving a third party. Using the technology without the requirement for the transaction to the participants for the central clearing authority. For triggering transaction delay, the character had trusted third-party intermediaries. There would be a faster reconciliation between participants and transacting parties without intermediaries. Additionally, BC solved financial transaction traceability concerns and security vulnerabilities by tracking the device interaction. This method was followed for transacting with other IoT devices. BC had a low chance of being hacked, and it improved data security.

In 2022 Li et al.²⁵ have implemented advanced federated learning for detecting False Data Injection Attacks (FDIA). By considering transformer models, the detector was displayed in edge nodes whereas the connection was established in the multi-head self-attention mechanism. In order to preserve data privacy, federated learning has the ability to train the model in the locally generated data, whereas it integrated with the Paillier cryptosystem. The simulation outcome of the developed model was experimented with With the help of IEEE 14-bus and 118-bus test systems, the experimental outcome was validated to maximize the performance.

In 2021 Liu et al.²⁶ have designed a deep reinforcement learning model for automatically transferring the Bitcoin representation with high frequency. However, the reinforcement technique has been validated with an advanced machine learning model for predicting the static price by applying with real-time Bitcoin, and also the transaction strategy was constructed by integrating with Proximal Policy Optimization (PPO) algorithm with LSTM to effectively construct policy. However, the various empirical studies were analyzed and validated to provide a better trading strategy to enhance the performance to effectively handle the high-frequency transactions in financial products.

Integrating Liu et al.'s approach of smart contracts in the blockchain financial system

For ensuring efficiency and scalability performance in blockchain-based financial security systems, the Liu et al.²⁶ strategy is suggested. This work has been mainly focused on enhancing the performance of Bitcoin trading strategy with the help of a deep reinforcement mechanism by integrating with the PPO algorithm. By leveraging this technique could boostup the training speed to ensure better decision-making process in complex and dynamic network environments especially in financial security systems. With the help of PPO-based DRL model could have the possibility to optimize the flow of smart contracts to reduce the usage of resources and transaction costs. During the execution process, the smart contracts were predefined by incorporating the reinforcement learning model based on analyzing the different behaviour in terms of dynamic environment and other external factors. Thus, it has the ability to optimize the decision-making process in smart contracts to enhance the performance for managing the risk to make the model more efficient and provide better scalable solutions. In blockchain financial security system model the consideration of PPO-based DRL framework could select the appropriate path by minimizing the transaction cost to ensure better security with the different blockchain systems.

Research gaps and challenges

The existing financial security model suffers from transparency issues. Also, it is not secure due to third-party services and struggles to manipulate data and spoofing-related issues. It gives poor execution. Sometimes, it may lead to data theft and privacy leakage. Therefore, the advantages and disadvantages are listed in Table 1. Consortium Per-Missioned Network¹⁷ easily reduces the inefficiency and complexity. Also, it reduces the disturbance to the stock exchange platform. So, it effectively trains the model. But, it takes longer to train the

Author [citation]	Techniques	Benefits	Issues
Abdallah et al. ¹⁷	Consortium Per-Missioned Network	It easily reduces inefficiency and complexity It reduces the disturbance to the stock exchange platform. So, it effectively trains the model	It takes a longer time to train the model It gives weak transparency
Tsoulias et al. ¹⁸	P2P NETWORK	It easily monitors and visualizes the changes in BC data It obtains reliable outcomes	It is hard to replicate the data and smart contract. Hence, it decreases performance It suffers from overfitting issues
Wang et al. ¹⁹	Edge computing	It effectively stores financial data It effectively monitors fraudulent transactions	It is hard to handle intensive computations It decreases the quality of service
Amponsah et al. ²⁰	ICN	It enhances the security of transactions in the blocks It gives promising results	It does not allow flexible authorization It struggles to achieve privacy preservation for data owners
Agyekum et al. ²¹	NHIS	It reduces the business cost It significantly minimizes the risk in finance	It suffers from data theft It gives poor data management
Shrivastava et al. ²²	CBDM	It improves corporate efficiency It gives high security, audit ability, privacy, and accountability for finance	It obtains a lack of transparency between the parties It takes a long line of communication
Li et al. ²³	Smart Contracts	It does not damage the information quality It enhances the throughput and latency of the model	The cost of the implementation is high It obtains low performance
Ahamad et al. ²⁴	Peer-to-peer	It gives high-integrity results It gives better data security, traceability, and immutability	It does not give reliable, secure storage for financial security It is hard to handle privacy

Table 1. Benefits and issues of blockchain-based financial security system.

model and gives weak transparency. P2P NETWORK¹⁸ easily monitors and visualizes the changes in BC data. Also, it obtains reliable outcomes. Yet, it is hard to replicate the data and smart contract. Hence, it decreases performance and suffers from overfitting issues. Edge computing¹⁹ effectively stores financial data. Also, effectively monitors fraudulent transactions. But, it is hard to handle intensive computations and decreases the service quality. ICN²⁰ enhances the security of block transactions and gives promising results. But, it does not allow flexible authorization, and also it struggle to achieve privacy preservation for data owners. NHIS²¹ reduces the business cost. Also, it greatly minimize the risks for all parties in the finance. Yet, it suffers from data theft and poor data management. CBDM²² improves corporate efficiency. Also, it gives high security, audit ability, privacy, and accountability for finance. Yet, it obtains a lack of transparency involved and takes a more time to communicate. Smart Contracts²³ do not damage the information quality. Also, it enhances the throughput and latency of the model. But, the cost of the implementation is high, and it obtains low performance. Peer-to-peer²⁴ gives high-integrity results. Also, it gives better data security, traceability, and immutability. Yet, it does not give reliable, secure financial security storage and is also hard to handle privacy. Hence, these issues motivate to implement BC-based financial security system with advanced deep learning techniques.

Blockchain-based financial security system: need of financial security and blockchain model

Need for financial security

Implementing BC solutions for financial services leads to various benefits for financial security. Financial security is the distributed consensus-based structure, which eliminates single points of failure and minimizes the intermediate like a messaging system, transfer agent, etc. Ethereum also enables the security to tamper-proof against fraud and third parties that are impossible to hack. BC in financial security also employs mutualized standards, shared processes, and protocols acting as a single shared source. By maintaining confidentiality and privacy, BC improves transparency, efficiency, and trust. Its hybrid and privacy network can handle hundreds of transactions per second.

Overview of blockchain

BC is demonstrated by the computer network, where all transactions are approved and recorded in a computer code chain. The transaction record detail is detailed in the public ledger, where the information's visible to everyone in the network. Hash signature is utilized to protect with authorized data which contain several blocks with the pointer in the form of the previous block hash. BC transactions in the network are broadcasted, and mining validates the process using special nodes in the network called miner nodes. After filling the block, the miner nodes append is hard as it needs subsequent blocks to maintain and modify the high-level security.

In the network, BC ledger is used to track the transaction and to replicate the transaction between the peer nodes. The identifiers or addresses know the public keys to maintain privacy efficiently by the network node. Some BC development, like Ethereum, follows protocols called smart contract. The computer protocol helps to enforce and verify the contract among the parties on BC based on the smart contract. Flow diagram for financial security using the block chain is defined in Fig. 1.

Types of blockchain

Many consensus mechanisms are used based on the BC technology and their implementation type. The major types of BC implementation are permissionless and permissioned.

Permissionless BC: It is the public implementation BC, where the users are nodes and the network is connected through the internet. Peer-to-Peer network (P2P) concept was used for the implementation. Here is

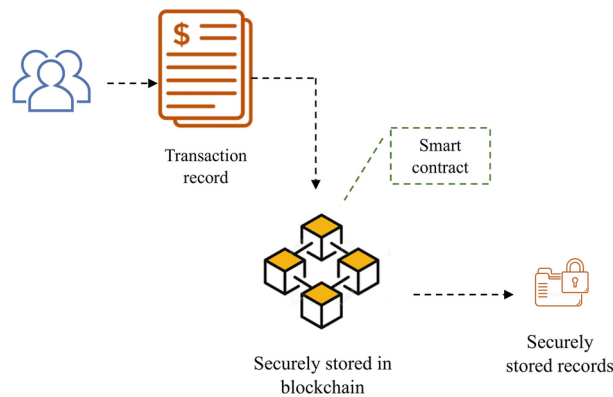


Fig. 1. Blockchain diagram for financial security.

the distributed architecture used by the P2P, where no clients are the administrator. Flat topology connects all the clients in the network, where every peer has the same rights to access the same resource.

Permissioned BC: The private BC is where the nodes are authenticated. For some implementations, the managing role responsibility is taken by the entity and allows them to access the data accordingly.

Development of a financial security system using smart contract in private ethereum consortium blockchain

Private ethereum consortium blockchain

PEC-BC²⁷ is used to predict the storage and prospective performance of the system to check whether the transaction reaches a certain level. World State is the core issue that affects the storage and performance of Ethereum. Ethereum is developed for both private and public BC. More voluntary nodes operate the public Ethereum, and the miners are used for block creation. Private Ethereum is designed to be accessed only for limited users. It is a more relaxed assumption than the public one. Here the consensus is attained with less computation, and predetermined nodes achieve the block creation. A consortium BC is also called permissioned BC, where the participant is limited in numbers. The hybrid architecture of the consortium BC features both public and private BC. In the participating members, every node is an individual organization. Based on pre-select participant size, the BC tends to determine these nodes. Through the gateway of the member institution, the consortium access the network. The consortium chain platform offers network transaction monitoring, information authentication, member management, and data read and write permission authentication. To access the ledger, permission is assigned to every member by the consortium. The major example of the consortium BC is the Hyperledger project. PEC-BC utilizes the transaction to achieve the contract with the help of transferring data. The performance and storage predicted by the PEC-BC resulting from the participant of authorized node, that helps to fix the smart contract with evaluable complexity is demonstrated by business framework and the business scale defined by the transaction volume. Architectural view of the private Ethereum consortium BC is shown in Fig. 2.

Considering the private ethereum consortium blockchain, the financial security system is further emphasized by adopting the permissioned nature and privacy features facilitates for maximizing the security and scalability performance. However, the permissioned nature of the financial transactions helps the blockchain facilitate to provide of authorized access, in which it maximizes security by considering the number of validators. It helps to avoid malicious activities to prevent from unauthorized access. The data leakage is effectively prevented in private ethereum consortium blockchain in financial transactions to secure the data accurately to minimize the risk of fraudulent activities. For efficient transaction processing, a smaller number of validators can ensure to increase the scalable performance compared to the public blockchain.

Challenges occur in terms of network centralization

Considering network centralization in blockchain, it contradicts the core principle whereas it leads to potential vulnerabilities like single point of failure. Thus, it effectively affects the transparency of the system model. Considering multiple financial transactions, the network congestion occurs during the peak loads. When it limits in transaction throughput, the network centralization can affect the overall robustness of the system performance. However, the constraints of block time remain challenging while causing a large number of delays in multiple transactions. During peak loads, it effectively affects the performance in the transaction throughput and system performance. If it hits with large number of concurrent request, the system becomes overloaded to provide poor query execution that mainly affects the performance of transaction throughput. Particularly, the peaks loads are effectively hinder the performance of private ethereum consortium blockchain for financial transactions. To resolve these issues, an effective strategy is adopted to provide scalable consensus mechanisms and suggest reliable performance.

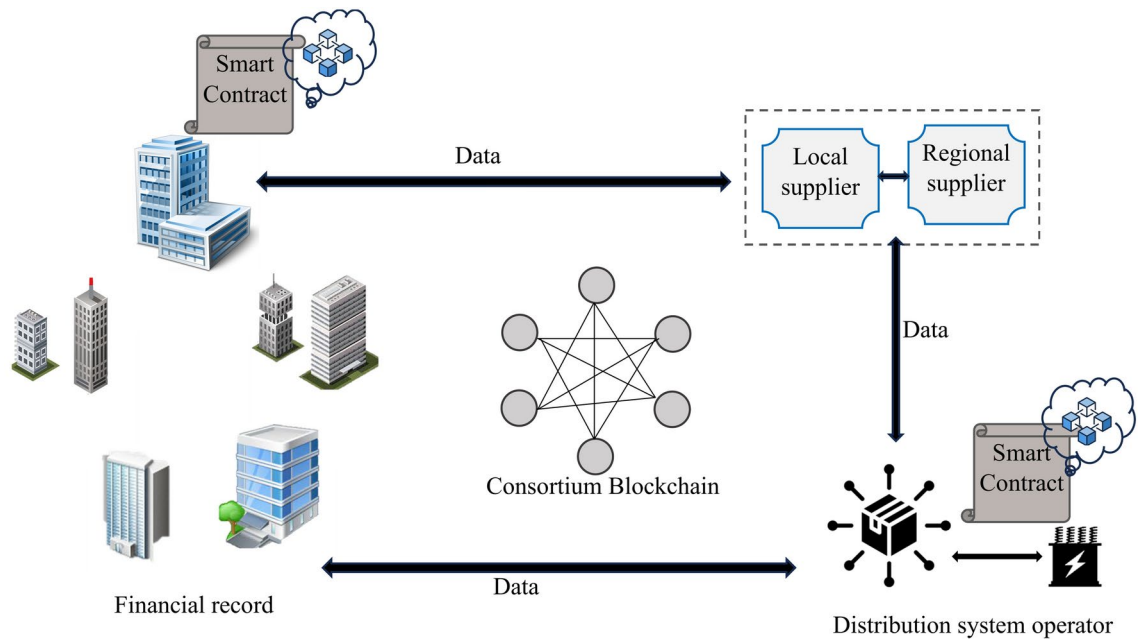


Fig. 2. Diagrammatic representation of Ethereum consortium blockchain.

Logic of smart contract

Smart contracts²⁸ are designed using Ethereum and contain computer programs in which miners verify execution outcomes. With the transaction of the users, execution and deployments are triggered. Every node participating in BC shows local virtual machine to execute smart contracts. The smart contract maintains and validates the globally shared state based on the transaction. The smart contract BC security level benefits from the BC properties such as integrity, transparency, availability, and security. There is the possibility to restrict the visibility for some consortium BC to the subset member of the contract in the same way as private transactions.

Proposed financial security system

The financial and governmental organizations combined to form a stock market platform for exchanging bonds, shares, and other securities in transactions called trade. The most trusted third party is needed for the online transaction due to the opaque information and digitization of currency in both parties to the transaction. Because of the share exchange and trade, it allow investors to be attracted. However, the weakness of over-reliance on intermediary loans has led to major disruptions in non-refundable services and day-to-day micropayments. In addition, there is a chance of personal information leakage and an increased transaction cost. Also, there have many limitations while including the intermediaries, such as the long time for financial settlements between broker and central bank, single point of failure, high transaction fees for brokers, and weak transparency. One of the main technologies to overcome the issues is BC technology. It comprises computer node networks share a common ledger without intermediaries. For the false transaction request, the Proof Authority (PoA) to evaluate transaction to cancel the claim. For a Virtual Power Plant (VPP), a Peer-to-Peer energy trading scheme is used for the smart contracts on the Ethereum BC. Distributed Ledger Technology (DLT) is implemented for the stock exchange system to separate the executing trades. Several consensus algorithms are implemented for the public BC network. An effective method is proposed to overcome such issues, shown in Fig. 3.

In this proposed framework, the development of a financial security system by the smart contract in the PEC-BC is used based on the deep learning method. At first, the customers financial data are collected and secured using BC. Here the optimization is done by the proposed DB-BOA inside the BC to select the leader block. With the additional new blocks, ADTCN with a consensus algorithm, is used to make smart contracts. Here, the parameters are optimized by the developed DB-BOA. The developed method performs well when comparing the developed ADTCN with other algorithms and conventional methods.

Dynamic butterfly-billiards optimization algorithm-based leader block selection on blockchain

Dynamic butterfly-billiards optimization algorithm

Here the implemented algorithm known as DB-BOA is used for the parameter optimization in developed ADTCN for security improvement. The optimization is designed by combining two algorithms such as DBOA and BOA. The advantage of using DBOA is that it can solve various problems with better results and avoid local optimal problems. BOA shows high level of exploitation, exploration, and balancing features. Although DBOA and BOA have better functions, they also have some issues. DBOA has a slightly high execution time and BOA. To solve this challenge, DB-BOA is developed, and the formulation is shown in Eq. (1)

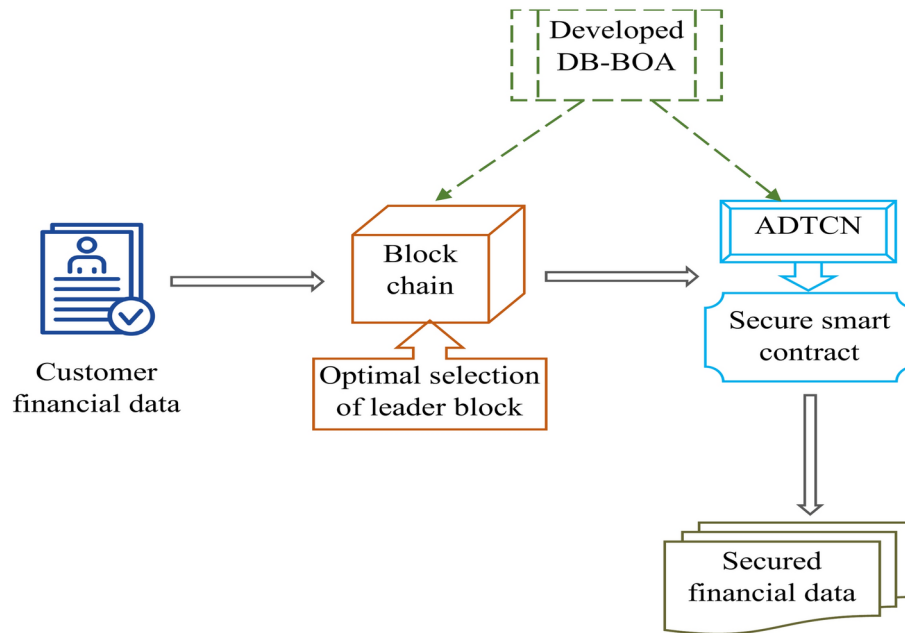


Fig. 3. Diagrammatic view of developed DB-BOA for financial security system.

$$rand < \left(\frac{bestfit}{worstfit} \right) \quad (1)$$

From Eq. (1), the optimal value is obtained from DBOA, or else it is obtained by BOA. The description of two existing algorithms is briefly explored as follows.

DBOA²⁹: is formed by the development of the Local Search Algorithm based on Mutation Operator (LSAM), which improves the exploitation phase of the Butterfly Optimization Algorithm (BFOA). DBOA algorithm calls the LSAM, which shows the best solution at each end of DBOA iteration.

BFOA: It is the optimized metaheuristic algorithm. This algorithm imitates the butterfly's behavior of finding a mating partner and food foraging. The chemoreceptors present in the butterfly's bodywork as sense receptors. These chemoreceptors are used for smelling or sensing foods or flowers. Also, the fragrance used in the BFOA algorithm guides the search agent movements. If the butterfly senses best fragrance butterfly, then moves towards particular butterfly is called exploration based on the intensity of the fragrance. If the fragrance is not sensed by any of the butterflies within the search space, the butterfly moves to new random places for local search or exploitation. Equation (2) shows the determination of the function of stimulus intensity defined by the fragrance in the BFOA algorithm.

$$g = dJ^b \quad (2)$$

here g does the butterfly emit the fragrance magnitude, d denotes sensory modality with range $[0, 1]$, J is the stimulus intensity of fragrance emitted by the butterfly, and b shows the sensory modality power exponent range $[0, 1]$. For updating the butterflies, there are two equations based on the value of g . Equation (3) represents the global search, and Eq. (4) represents the local search.

$$y_j^{u+1} = y_j^u + (s^2 \times h^* - y_j^u) \times gj \quad (3)$$

$$y_j^{u+1} = y_j^u + (s^2 \times y_k^u - y_i^u) \times gj \quad (4)$$

here h^* indicate the current iteration's best solution, gj is the j th butterfly fragrance magnitude, s indicate the random value at an interval $[0, 1]$, y_k^u indicate the k th butterfly, and y_i^u indicate the available solution space.

There are two phases in the BFOA algorithm like local search and global search. In this algorithm, the switch probability ρ is controlled between two phases, which means the value of s smaller than ρ it will perform global search. Otherwise, it will moves towards local search.

LSAM: It works under two principles one is improving best solution to enhance the quality of other solutions. The current best solution h^* at the end of each iteration from BOA is received first at LSAM. Then the mutation process is applied to c . If the mutation process is fitter h^* , then it is replaced by the new solution. Or else, a random solution is selected by the LSAM from the remaining solution. Here, the selected solution is replaced

with a new mutated solution. LSAM repeats the steps Num_itr times. If it finds a better solution, it assigns to h^* .

BOA³⁰: Billiards is the most interesting and favored game for many people. The intelligent activity the player performs is the strategy to design the optimizer. BOA is based on the player's behavioral impression of the game.

BOA is the metaheuristic algorithm, a population-based approach where the billiards balls are the members. The candidate solution is the BOA member based on the position of the value of the problem variable assigned. Equation (5) shows the mathematically modeled BOA members using a matrix.

$$Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_j \\ \vdots \\ Y_N \end{bmatrix}_{O \times n} = \begin{bmatrix} y_{1,1} & \cdots & y_{1,e} & \cdots & y_{1,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ y_{i,1} & \cdots & y_{i,d} & \cdots & y_{i,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ y_{M,1} & \cdots & y_{M,d} & \cdots & y_{M,n} \end{bmatrix}_{O \times n} \quad (5)$$

Equation (6) shows random initialization of BOA members for implementation process.

$$Y_j : y_{j,e} = mc_e + s \times (vc_e - mc_e) \quad (6)$$

here Y is the population matrix of BOA, Y_j represents the j th candidate solution, $y_{j,e}$ indicates the value of e th variable suggested by the j th population member. The Number of BOA members represent by O , a number of variables represent n , the random number s in the interval $[0 - 1]$ mc_e and vc_e are the lower and upper bound of the e th variable.

The candidate solution is each population member, and the value of the objective function calculates it. The objective function value represented by the vectors shown in Eq. (7)

$$G = \begin{bmatrix} G_1 \\ \vdots \\ G_j \\ \vdots \\ G_O \end{bmatrix}_{O \times 1} = \begin{bmatrix} G(Y_1) \\ \vdots \\ G(Y_j) \\ \vdots \\ G(Y_O) \end{bmatrix}_{O \times 1} \quad (7)$$

G is the vector value of the objective function and G_j indicates specified value for j th the population member's objective function. The population member's updated position in each iteration evaluates the objective function's new value. With the help of this iterative process, the population is upgraded in this algorithm.

In BOA, the population members are updated by player's behaviour in placing the balls in the pocket. The billiards table has 8 pockets; each pocket is considered a better member among the population. Here, the pocket is chosen at random in the BOA design. Equation (8) shows the calculation of the proposed random position for each member of the population.

$$Y_j^{new} : y_{j,e}^{new} = y_{j,e} + s \times (Q_{j,e} - J \times y_{j,e}) \quad (8)$$

Then the new position is improved for the objective value function and Eq. (9) shows the previous position replaced by the new position.

$$Y_j = \begin{cases} Y_j^{new}, & G_j^{new} < G_j \\ Y_j, & else, \end{cases} \quad (9)$$

here Y_j^{new} is the evaluation of current state of j th the population member $y_{j,e}^{new}$ represents its e th dimension. G_j^{new} Indicates the objective function value, Q_j is the chosen pocket for the population member, $Q_{j,e}$ indicates e th a dimension s is a random number in the interval $[0 - 1]$, and J is a random number selected randomly from set $\{1, 2\}$. Figure 4 depicts the flow chart for the recommended model and the pseudocode is given below.

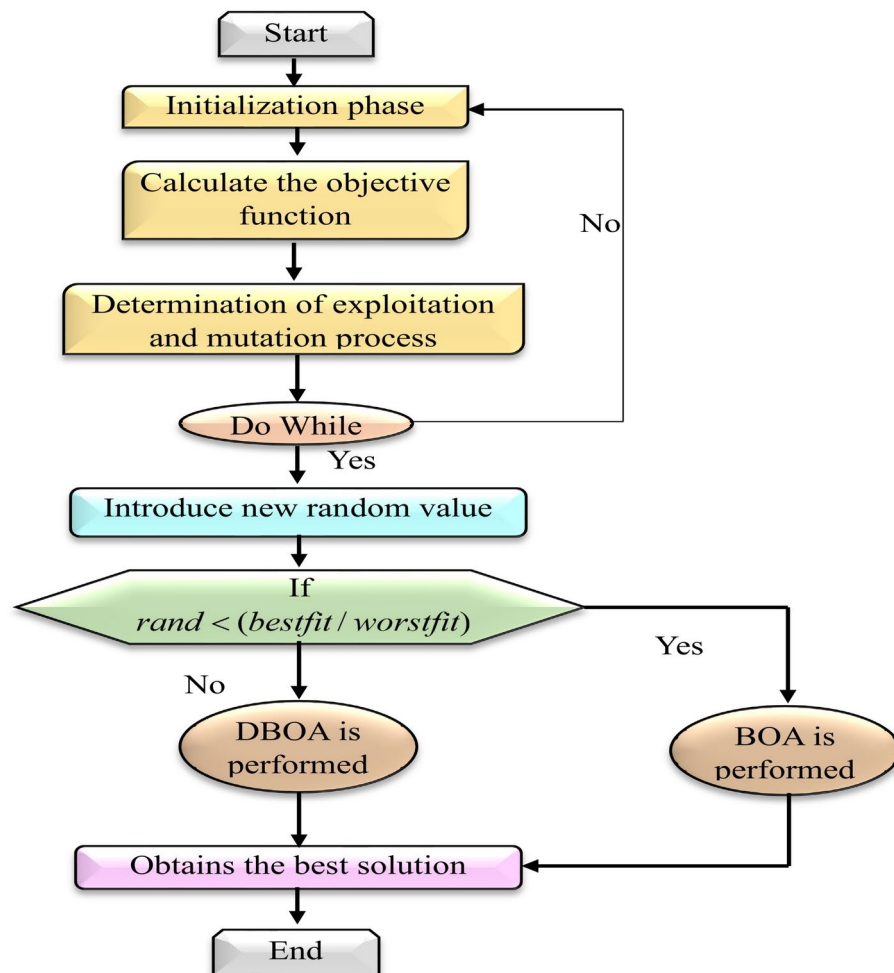


Fig. 4. Flow diagram for developed DB-BOA.


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Initialize the parameter of the DBOA and BOA
Determine population and iteration value
Evaluate random parameter as rand.
While iteration begins
  If rand < (bestfit / worstfit)
    DBOA is performed
    Initialize butterflies' population position.
    Update exploration stage in Eq. (3)
    Upgrade exploitation phase in Eq. (4)
    Update the best global solution
  Else
    BOA is performed
    Determine BOA member position in Eq. (6)
    Determine the new status of  $j^{th}$  candidate
    solution in Eq. (8)
    Update the  $j^{th}$  candidate solution in Eq. (9)
    Best candidate solution
End while
Returns optimal solution

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Algorithm. DB-BOA

Leader block selection

Leader blocks³¹ are selected by the BC features with the highest reputation value in the reputation group. In leader selection process, the main category are security and incentives. The high reputational value nodes are responsible for the system security. In the round of consensus, the leader will get more rewards. The leader block helps to broadcast the transaction block together with the signature. The selection of the leaders increases the throughput of the protocol. By collecting the legitimate transaction, the leader creates a transaction block. The leader block selection also has some limitations. The leader selection schedule is predictable and allows one to attack the next leader. Due to the high communication overhead and latency of multiple, there may be Low network scalability. The objective parameters are used in the proposed DB-BOA algorithm to solve this issue. By introducing the newly derived strategy of the DB-BOA algorithm, the leader block is selected optimally using several advanced features in the optimization algorithm. In financial security systems, the high frequency of transactions is effectively addressed with the help of developed DB-BOA algorithm, especially in consortium blockchain. However, optimizing the leader block and minimization of consensus latency becomes the crucial task to maximize the scalability and reliability of the system. In particular, the leader block is selected with the help of choosing the number of validators, in which it effectively generated in the next block. However, this selection process helps to consistent performance and avoid unnecessary delays in the system. With the help of butterfly strategy and billiard approach helps the recommended method to be more focused to reduce the consensus latency with the help of different number of validators. By ensuring its adaptability and efficiency in DBBOA algorithm, it tends to provide high throughput performance and minimize latency consensus in high-frequency transaction environments.

The objective function value for optimal selection of the leader block is shown in Eq. (10)

$$Obf1 = \arg \min_{\{Lb^{bc}\}} [CT + CC + MS] \quad (10)$$

here Lb^{bc} represents the leader block in the BC. CT is the computation time, CC is the communication cost, and MS is the memory size. Comparatively, computational time, communication cost, and memory size are demonstrated below.

Computational time: It demonstrates whether an algorithm is efficient in running within an acceptable time frame.

Communication cost: It measures the quantity data sent between the communication parties.

Memory size: The storage capacity of carrying information between the data defines the memory size. The solution encoding framework for the leader block selection is shown in Fig. 5.

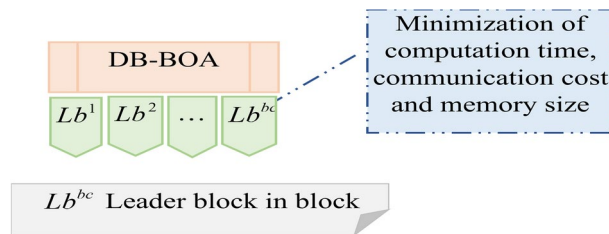


Fig. 5. Schematic representation solution encoding for leader block selection using developed DB-BOA.

Adaptive deep temporal context networks for improving security in blockchain Block security validation by ADTCN

Comparison to existing smart contract verification methodologies with proposed model

For detecting security vulnerabilities in smart contracts, the hybrid attention mechanism (HAM) has been employed³². In blockchain technology, the smart contract runs as a special program, which is incorporated without relying on a third party. By combining the active and semi-supervised BERT model, the smart contract is performed in the blockchain technology³³. This model ensures to provide better generalization ability with the help of encoder representation in the transformer networks to select the accurate code from the unlabeled datasets. Utilizing deep learning techniques in smart contracts³⁴, it helps to provide the benefits of immutability, and security in the blockchain. In the Ethereum blockchain, the Finite State Machine (FSM) models are adopted to verify the composite smart contract³⁵. However, the consideration of these models shows several challenges are listed below. In traditional techniques, the increasing number of smart contracts shows unrealistic performance, in which it is difficult to understand the complex patterns. It is vulnerable to provide better security measures in the conventional models. Due to insufficient labeled data, the smart contract is not effective in blockchain model whereas it requires more training time. Security breaches are often occurs in the Ethereum blockchain while pursuing smart contracts. Thus, it leads to huge information loss and vulnerable to attacks.

On the other hand, the ADTCN model is adopted in this research work to ensure better security smart contracts in financial system. The consideration of DTCN model can effectively analyze the time series data to detect anomalies in order to learn complex patterns within sequential information. Unlike traditional techniques, the DTCN model effectively works to capture the long term dependency pattern in time series data. Especially this model helps to analyze the financial transaction data to detect fraudulent patterns. In order to enrich the model performance, the DTCN is further improved by integrating the adaptive concept and named as ADTCN model. ADTCN model facilitates to show better performance to provide security in the financial systems and make the model more effective to provide reliable performance. ADTCN is the deep prediction framework, which has the components of Multi-modal Joint Embedding (MJE), Temporal Context Learning (TCL), and Multiple Time-scale Temporal Attention (MTTA). The whole network was trained by these components optimized by the DB-BOA algorithm. Here the optimal selection of leader blocks in BC trained by the ADTCN network with the parameter optimized by the developed DB-BOA to achieve smart contract. The layers utilized in the ADTCN model are listed below.

MJE: It is the first phase of ADTCN to produce the unified deep indication of multi-modal sequence data. The user data interaction mostly coincides with individual popularity. This method is not only for knowing about data appearance but also for identifying the base of the content. For the embedding network, the input data are the bi-modal sequence data. For the visual content parametrize, the data sequence influences the user and the two stream Feed Forward Neural Network (FNN) analyzes both the user pipeline and photo pipeline. The trained ResNet starts the photo analysis pipeline to produce high-level dimension representation. At the same time, a group of features was adopted to identify the user influence average value of view in the user analysis pipeline in a user group. Every pipeline in the embedding network consists of four layers, where the two hidden layers splits the same dimension of the hidden neural node. The original space non-linear mapping is performed by new latent space. For embedding the network, the activation function is applied. The neural network dropout strategy has been adopted to protect overfitting in the training phase. The final layer of the two-stream FNN generates the end of architecture, which is contaminated to reduce embedding loss.

TCL: This is the learning phase, where the temporal and sequential coherence are learned from the TCL prediction. It correlates with the information of time and it is different from traditional learning of prediction targets. Building the temporal context is the first step for every future popularity. The unregular earlier data sequence built the PTC, which is suitable for denoting periodic popularity patterns in the long term. The prediction target of NTC from the old data sequence is applied to demonstrate fluctuation popularity in the short-term range. Here the designed PTC and NTC determine the corresponding elements in the temporal context.

MTTA: Every contextual dynamic impact was considered the item into prediction. The temporal attention along multiple timescales is for new popularity prediction between the previous and current instances. The multiple timescales infer by the attention mechanism from both the multi-scale of time series and the weight of relatively hidden learned the multi-time scale coherent data of the time series.

Resilience and robustness of financial security systems in high-frequency blockchain

Utilizing smart contract in financial system helps to improve the resilience and robustness by adopting with developed ADTCN model. Here, the leader block is selected using the developed DB-BOA algorithm by considering different number of validators. Breaking the transaction data into smaller portion meticulously reduce the transaction latency by leveraging the consensus mechanism in the ADTCN model. Integrating deep learning model in blockchain can effectively enhance the decision making performance and ensure trustworthy performance to secure the financial data. Under heavy load network, the authenticated data is further transferred to process the data by reducing the malicious activities in the network. Based on heavy network condition, the developed ADTCN model progress only the relevant data thus, it can reduce the transactional latency. Focusing the number of validators in blockchain ensures to reduce the transactional latency in different network load. This can be effectively works when the transaction is effectively progressed without affecting its security in order to prevent from malicious activities. Also, the transactional latency is highly minimized by considering the consensus algorithm. Here, the consensus algorithm is performed with the help of ADTCN model to maintain the security in the financial data under different network load. By minimizing the network congestion to improve the security, the ADTCN model is leveraged to reduce transactional latency in blockchain. Optimizing the leader block in blockchain using DB-BOA algorithm facilitates to improve the consensus stability under heavy network load. Utilizing consensus mechanism in ADTCN model can handle the large volume of transaction data to enhance the stability of the model. Minimizing transactional latency and maximizing the consensus stability helps to ensure better resilience and robustness performance in the developed financial security system by leveraging the consensus algorithm with ADTCN model.

Proposed ADTCN

ADTCN incorporates both temporal context and temporal attention. The embedding of the collaborative network was performed efficiently, and context was applied for the securing purpose. ADTCN is recommended to analyze new popularity with temporal coherence on multiple time scales. The developed DB-BOA model optimally determined the secure smart contract. The objective measures of this process are formulated in Eq. (11)

$$Obf2 = \arg \max_{\{Hn^D, Ep^D, Se^D\}} \left[(Acc + Pre + NPV + MCC) + \frac{1}{FPR} \right] \quad (11)$$

here Hn^D is the hidden neuron count in DTCN, Ep^D is the epoch count in DTCN and Se^D is the steps per epoch in DTCN which ranges from $[5 - 255]$, $[5 - 50]$ and $[50 - 250]$. Acc is the accuracy, Pre is the precision, NPV is the Negative Predictive Value, MCC is the Matthews correlation coefficient and FPR is the False Positive Rate respectively.

Accuracy: it is used to measure the performance of the model shown in Eq. (12)

$$Acc = \frac{Ta^p + Ta^n}{Ta^p + Ta^n + Fb^p + Fb^n} \quad (12)$$

Precision: it is used to measure the accuracy of positive prediction shown in Eq. (13)

$$Pre = \frac{Ta^p}{Ta^p + Fb^p} \quad (13)$$

NPV: It is the probability of a person not having a problem who is tested negative equated in Eq. (14)

$$NPV = \frac{Ta^n}{Ta^n + Fb^n} \quad (14)$$

MCC: It is the changes between actual and predicted value computed in Eq. (15).

$$MCC = \frac{Ta^p \times Ta^n - Fb^p \times Fb^n}{\sqrt{(Ta^p + Fb^p)(Ta^p + Fb^n)(Ta^n + Fb^p)(Ta^n + Fb^n)}} \quad (15)$$

FPR: It is the probability of evaluating the negative cases as positive and the total amount of negative events in Eq. (16)

$$FPR = \frac{Fb^p}{Fb^p + Ta^n} \quad (16)$$

here Ta^p is the true positive, Ta^n is the true negative, Fb^p is the false positive and Fb^n is the false negative. The diagrammatic view of the proposed ADTCN for secure smart contract explained in Fig. 6.

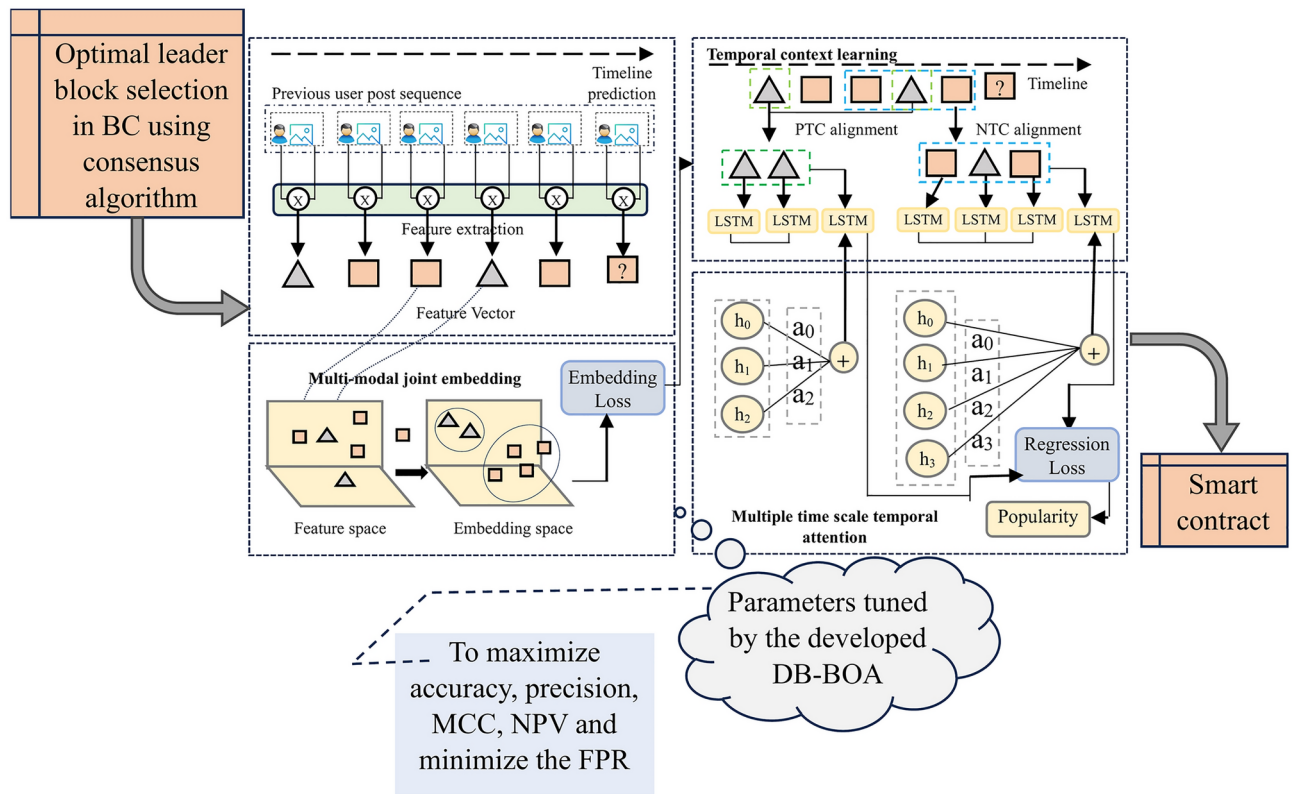


Fig. 6. ADTCN for secure financial data using DB-BOA.

Results and discussions

Simulation settings

The developed ADTCN-based financial security system was executed in MATLAB 2020a, and the analysis were carried out. Several optimization models such as Mine Blast Optimization (MBO)³⁶, Water Strider Algorithm (WSA)³⁷, Dynamic Butterfly Optimization Algorithm (DBOA)²⁹, Billiards Optimization Algorithm (BOA)³⁰, Layered Gossip Protocol (LGP)¹⁹ and Gossip Protocol (GP)¹⁹ were utilized. Similarly, some of the classifiers such as Efficientnet³⁸, Resnet³⁹, Densenet⁴⁰ and DTCN⁴¹ were adopted.

Performance metrics

Sensitivity: It demonstrates the minimum absolute amount of variation identified by the factor determined in Eq. (17)

$$Sen = \frac{Ta^p}{Ta^p + Fb^n} \quad (17)$$

FNR is determined in Eq. (18)

$$FNR = \frac{Fb^n}{Fb^n + Ta^p} \quad (18)$$

Specificity: The analysis of the likelihood of the negative rate is explained in Eq. (19)

$$Spe = \frac{Ta^p}{Ta^n + Fb^p} \quad (19)$$

FDR: It is the measurement of both FP and TP is equated in Eq. (20)

$$FDR = \frac{Fb^p}{Fb^p + Ta^p} \quad (20)$$

F1-Score: It is used to determine the overall performance of the classification model shown in Eq. (21)

$$F1S = 2 * \frac{Ta^p * Ta^n}{Ta^p + Ta^n} \quad (21)$$

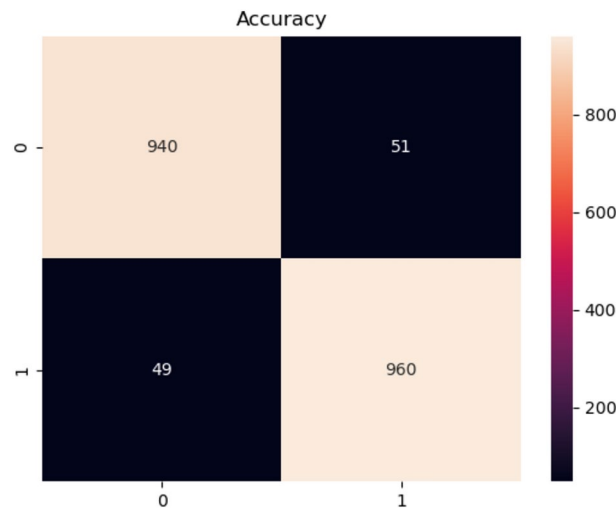


Fig. 7. Confusion matrix analysis for the financial security system based on the smart contract of the recommended model.

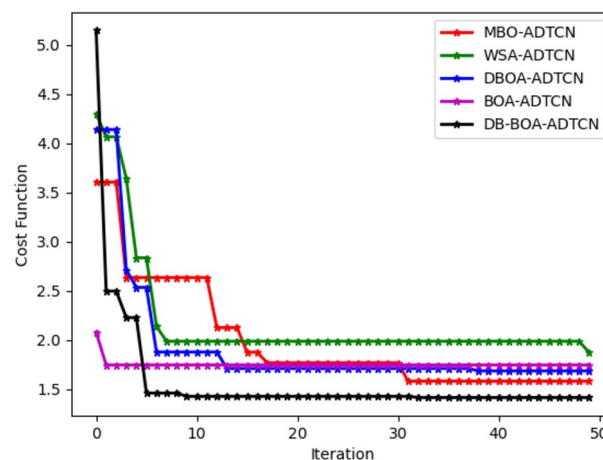


Fig. 8. Analysis of cost function of the developed model in financial security system.

Performance examination of the confusion matrix analysis

The confusion matrix measures the developed system shown in Fig. 7. Thus, it shows the actual and predicted value, which are tuned to analyze the accurate value. Each column contains two measures, 0 and 1. From the graph, the developed system has better accuracy, which assures the method's effectiveness.

Validation of proposed optimization algorithm in terms of convergence analysis

Figure 8 displays the cost function analysis of the proposed method with various algorithms. For validating the outcomes, the iteration count vs cost function varies from 0 to 50 and 1.5 to 5. While analyzing the cost function analysis, the suggested method reduces by 72% of MBO-ADTCN, 80% of WSA-ADTCN, 81% of DBOA-ADTCN and 82% of BOA-ADTCN, respectively. Hence the convergence of the developed method has enhanced to provide optimal outcomes.

ROC analysis in financial security system

ROC is referred to as Receiver-Operating Characteristics. Figure 9 shows the ROC varied with other classical models. This model is tuned with a rates of false positive as well as true positive. The developed model improved by 20% of Efficientnet, 24% of Resnet, 25% of Densenet and 29% of DTCN by taking a false positive rate is 0.2. Thus, the proposed method proved the effective outcomes.

Activation function-based analysis of the suggested framework

Figures 10 and 11 show the activation function-based comparative analysis of the suggested method with diverse algorithms and classifiers. Figure 10a elaborates the accuracy measurement for the algorithm. The accuracy value varies from 86 to 96. Considering accuracy of the model, the developed model shows 89% of MBO-

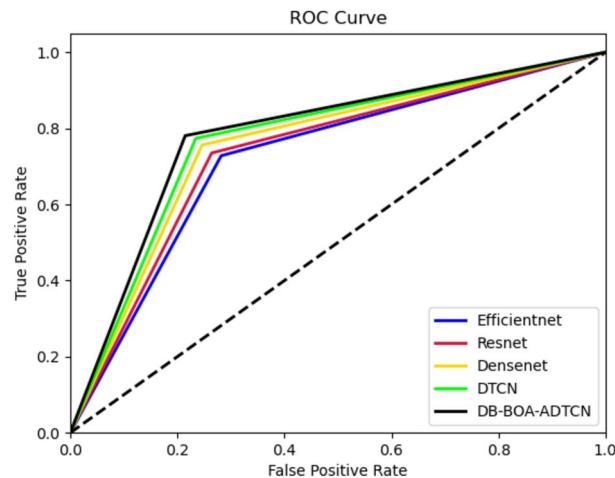


Fig. 9. ROC curve examination of the developed model for the financial security.

ADTCN, 92% of WSA-ADTCN, 90% of DBOA-ADTCN and 94% of BOA-ADTCN respectively focusing on TanH measure.

Block propagation delay analysis in financial security system

Figure 12 elaborates the evaluation of the block propagation delay for the algorithm with diverse models. Figure 12a block propagation value varies from 2000 to 4500. The block size value ranges from 5000 to 14,000. The value for the suggested method is enhanced by 68% of LGP with $\alpha=0.1$, 54% of LGP with $\alpha=0.3$, 60% of Block in GP with $\alpha=0.1$ and 57% of BOA-ADTCN at the block size value 7000.

Computational time analysis for the suggested method

Figure 13 displays the computation time validation for the recommended model contrasting with several algorithms. Figure 13a number of users varies from 5 to 40. The developed model minimized by 20% of LGP with $\alpha=0.1$, 10% of LGP with $\alpha=0.3$, 28% of Block in GP with $\alpha=0.1$ and 29% of BOA-ADTCN while taking the number of users as 10. During the implementation process, the developed model shows better computational time by validating with different number of users.

Throughput analysis for the implemented model

The throughput analysis of the recommended technique is compared over diverse optimized approaches is given in Fig. 14. The throughput value lies within the interval of 0 to 400. Figure 14a throughput value of the implemented framework is maximized by 30% of LGP with $\alpha=0.1$, 50% of LGP with $\alpha=0.3$, 5% of Block in GP with $\alpha=0.1$ and 85% of BOA-ADTCN, respectively, by considering the sending rate as 200. Hence, it is proved that the proposed approach has a high throughput value.

Execution time analysis for the suggested model over various optimized models

The proposed method of execution time analysis with various models is shown in Fig. 15. Here, the functionality measures of the designed model are estimated with the transaction varies from 0 to 17.5. The transaction of the implemented model is improved by 43% of LGP with $\alpha=0.1$, 39% of LGP with $\alpha=0.3$, 17% of Block in GP with $\alpha=0.1$ and 49% of BOA-ADTCN by considering the block size as 200. This shows the effective performance of the developed model.

Latency analysis for the suggested model over various optimized models

The proposed method for latency evaluation with various numbers of validators is shown in Fig. 16. Here, the functionality measures of the designed model are estimated with the number of validators varying from 1 to 20. The latency of the implemented model is decreased by 36% by 100 tx/sec, 7% by 200 tx/sec, 2% by 300 tx/sec, 5% by 350 tx/sec and 26% by 400 tx/sec by considering the validators as 5. This revealed the enhanced performance of the proposed method.

Statistical evaluation of the designed approach

Table 2 shows the comparative evaluation of the proposed ADTCN model over multiple algorithms. The developed DB-BOA has 11.7% of MBO-ADTCN, 32.2% of WSA-ADTCN, 19.1% of DBOA-ADTCN and 23.3% of BOA-ADTCN considering best measures.

Performance analysis of the developed DB-BOA with ADTCN for the financial security

Tables 3 and 4 illustrates the overall analysis of the financial security system compared with other classical methodologies. In Table 3, the accuracy is acquired as by 7.3% of MBO-ADTCN, 3.7% of WSA-ADTCN, 5.7% of

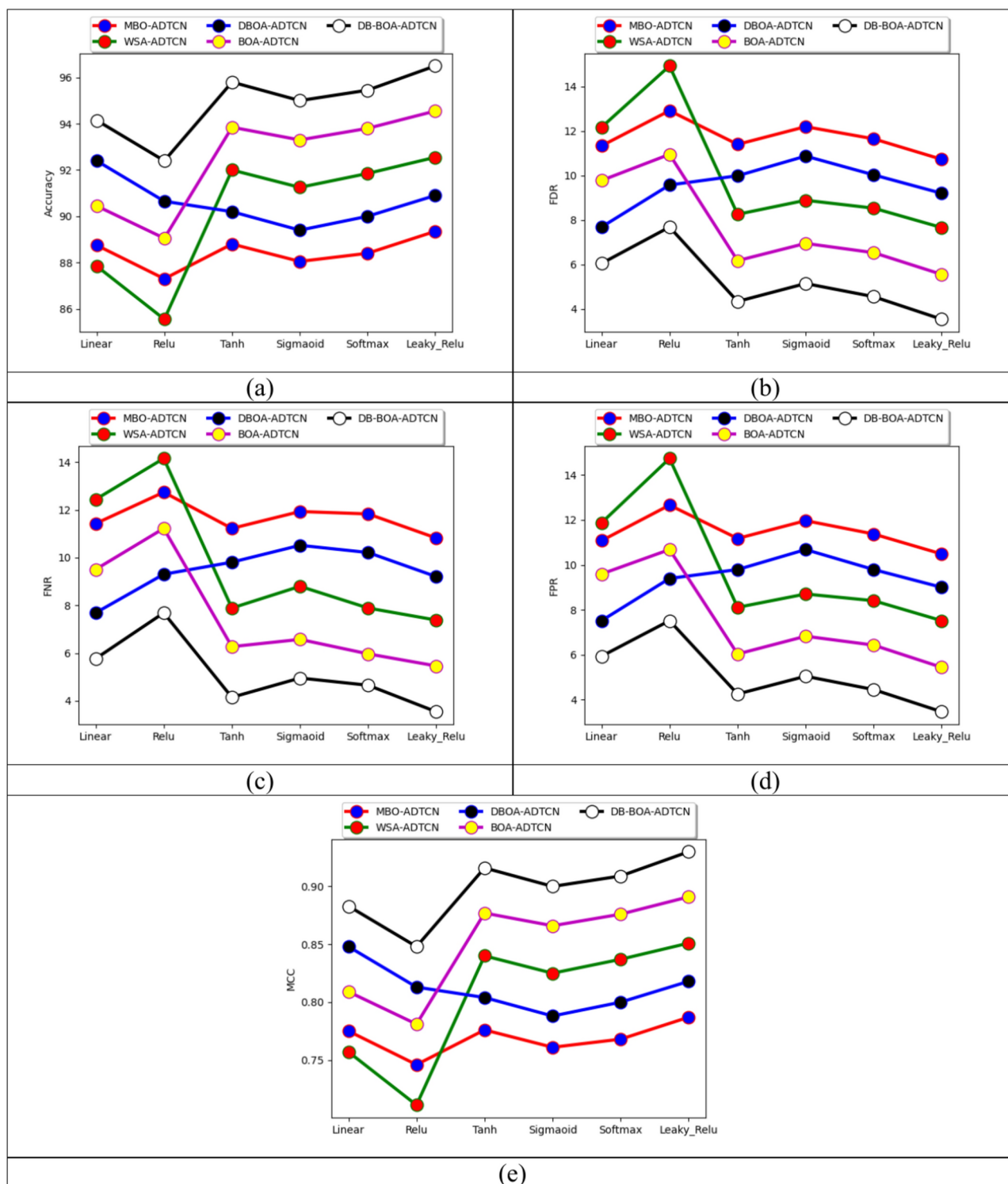


Fig. 10. Comparative evaluation of proposed financial security system using BC compared with multiple traditional algorithms for evaluating the (a) accuracy, (b) FDR, (c) FNR, (d) FPR, (e) MCC.

DBOA-ADTCN and 1.7% of BOA-ADTCN, which is lesser than DB-BOA-ADTCN. Hence the higher accuracy is used for the security system.

Scalability analysis of the developed model

The scalability analysis of the developed model in a financial security system is shown in Fig. 17. The scalability analysis in financial system ensures to provide robust performance, in which to enhance the performance with faster training times. By leveraging the large amount of data, the developed model facilitates to enhance the

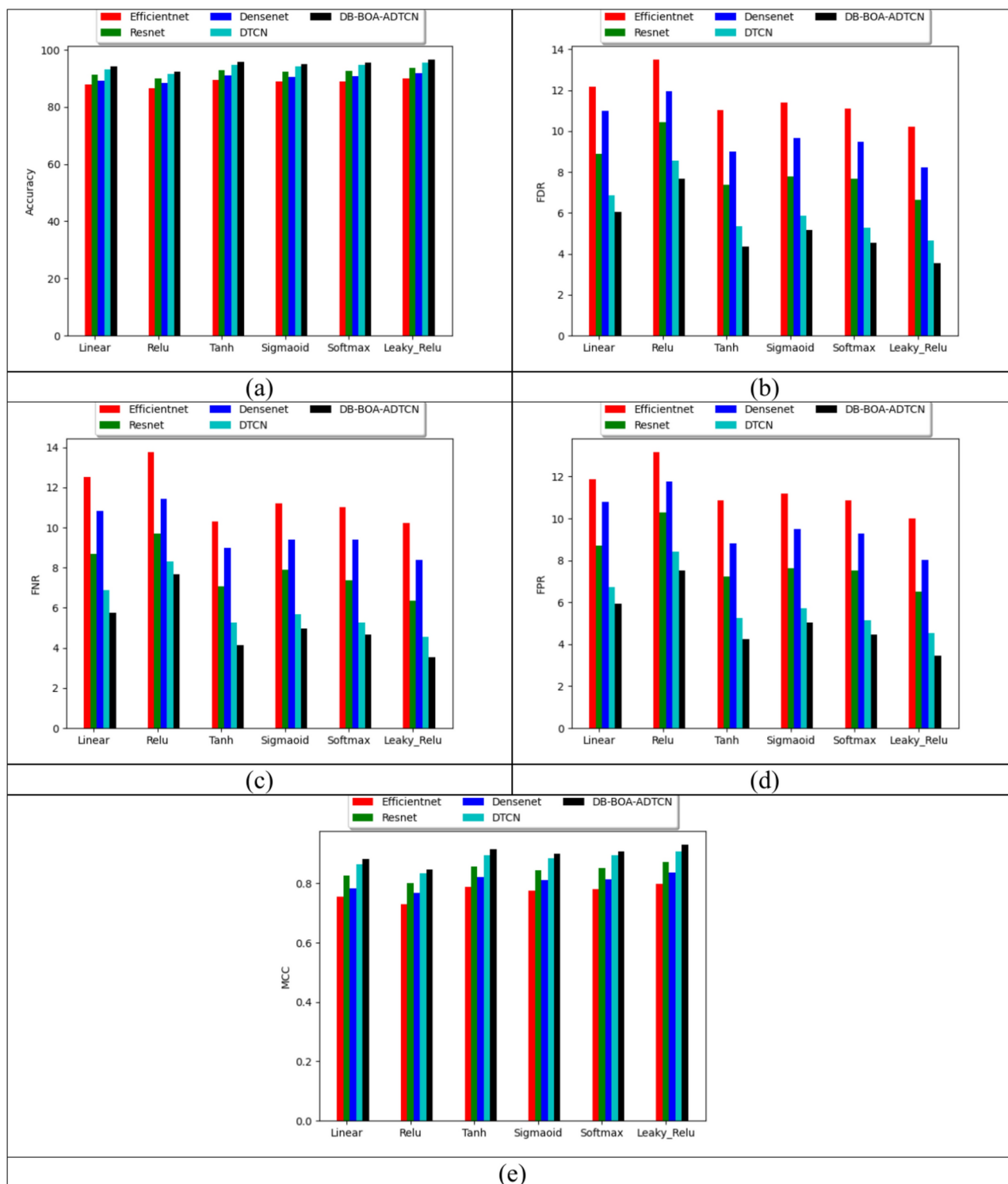


Fig. 11. Comparative evaluation of proposed financial security system using BC compared with multiple traditional classifiers for evaluating the (a) accuracy, (b) FDR, (c) FNR, (d) FPR, (e) MCC.

accuracy of the model without compromising the system performance. By increasing the scalable performance, the latency gets effectively reduced to identify the bottlenecks and optimize the performance by considering the number of validators. It is crucial to analyze the fast and reliable transaction that tends to enhance the user experience and provides better decision making process to maximize the overall system performance. Lower latency helps the model to effectively handle a large number of transactions in an effective manner.

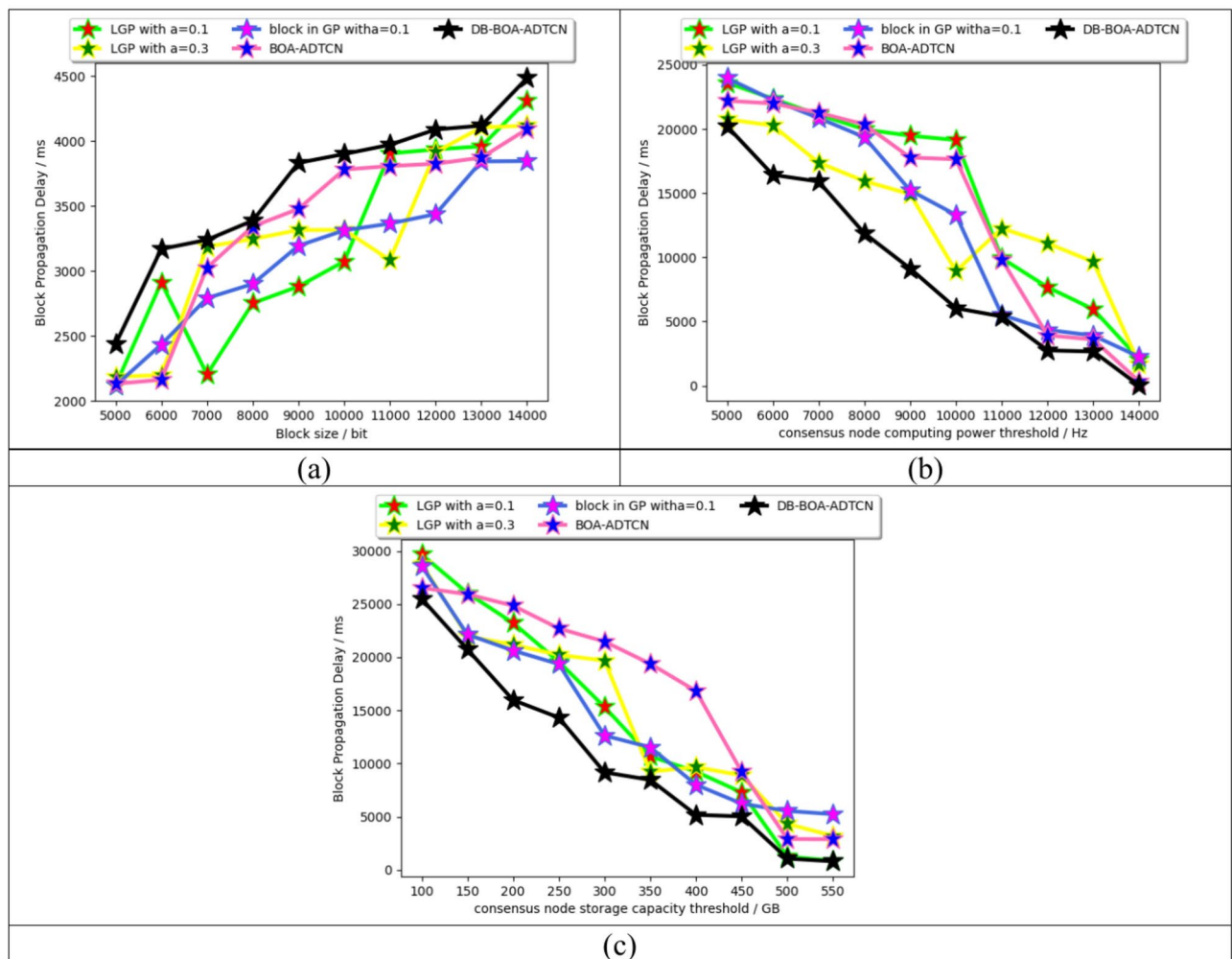


Fig. 12. Block propagation delay evaluation of suggested financial security system using BC compared with a traditional algorithm in terms of (a) Blocksize, (b) Consensus node computing power threshold, (c) Consensus node storage capacity threshold.

Computation time of the developed model

Table 5 elucidates the computational time of the developed DB-BOA algorithm and proposed ADTCN model by comparing with conventional techniques. Here, the developed DB-BOA-ADTCN model shows better processing time compared than the existing approaches. By evaluating the computational time, the developed model has the capability to identify the potential bottlenecks in the financial security system. Ensuring this performance, the developed model can minimize the delay while maximizing the model performance with fast transaction processing. Comparing the traditional techniques, the developed DB-BOA-ADTCN model achieves 13.003350 secs makes the model to attain promising and reliable outcomes.

Balanced trade-off between transaction speed and computational complexity

The consideration of the developed ADTCN model and DB-BOA algorithm facilitates to provide better balance among the transaction speed and computational complexity. For large-scale financial transactions, the developed model can process the financial data in a various format to minimize the latency and boostup the system performance, in which it handles a large number of transactions to manage the computational efficiency. The adoption of deep learning model leverages better trade-off among the transaction speed and computational complexity to estimate the model predictions and also processing time to run the model. Through its adaptive nature in the ADTCN model, it facilitates to process a large amount of financial data. Also, the developed model has the ability to select the appropriate features for reducing repeated computations to ensure a balanced trade-off between the transaction speed and computational complexity to maximize the scalable performance in order to meet the requirements of the financial security system. On the other hand, the computational complexity of the developed model is shown in Table 6. Here, the number of population and maximum number of iteration is defined as N_{pop} and $Maxiter$. Also, chromosome length is denoted as $Chlen$.

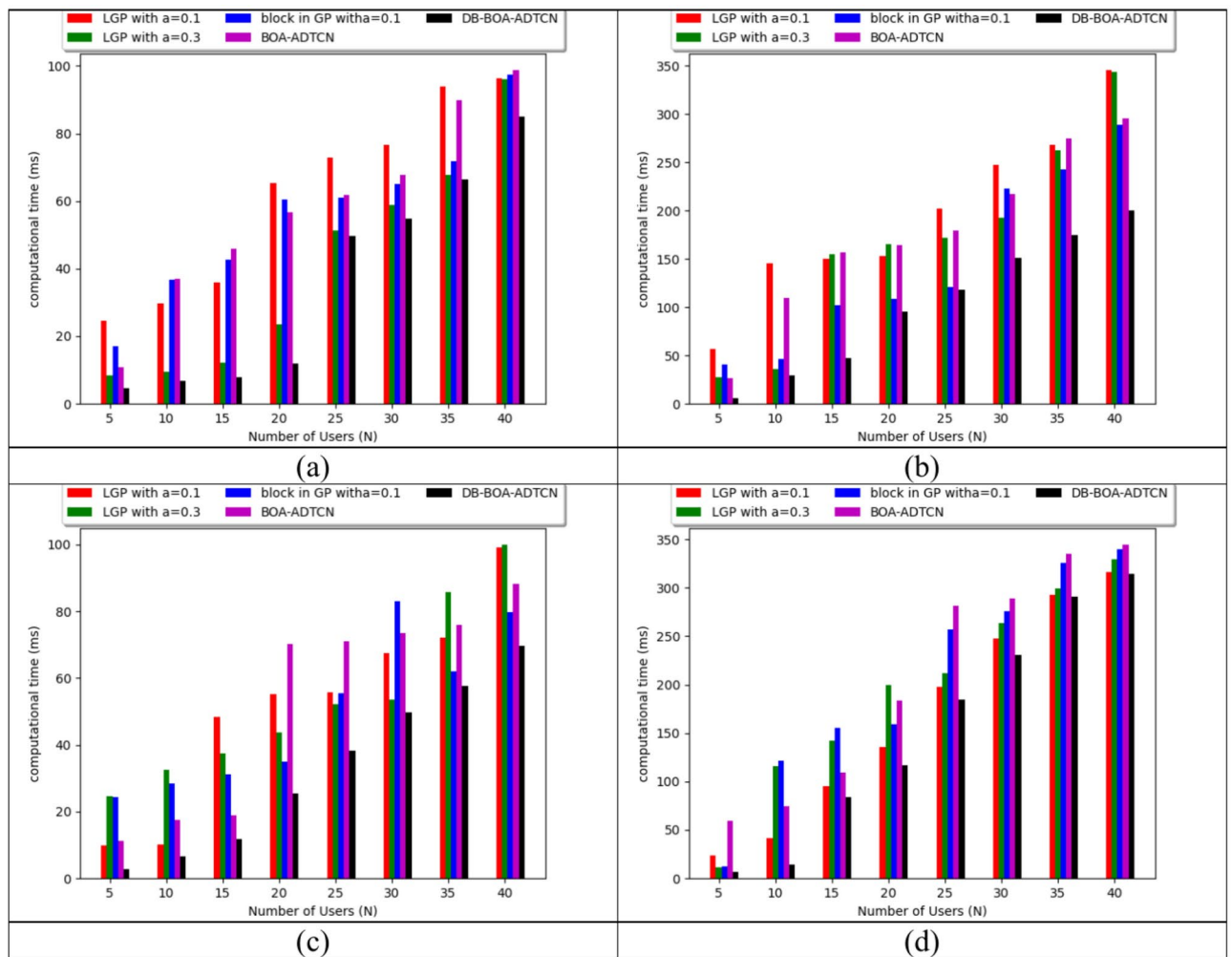


Fig. 13. Computation time of the proposed method in terms of (a) Decryption, (b) Encryption, (c) Re-decryption, (d) Re-encryption.

Robustness analysis of the developed model in financial security system

The robustness analysis is validated by considering the various techniques to prove the effectiveness in the developed model shown in Fig. 18. The robustness analysis is conducted by focusing on the different variations of noise coefficient like 5, 10, 15, 20, and 25. However, the developed model facilitates to detect the fraudulent activities to maintain the accuracy and reliability of the model to provide better security measures for preventing from data breaches. Robustness analysis of the developed model ensures to test across various conditions to show effective performance in financial security systems.

Throughput analysis of the developed model

Figure 19 shows the throughput analysis of the developed model in the financial security system by comparing with other existing techniques. By validating the throughput analysis, it helps the developed model to effectively handle the larger volume of data in financial security system by considering existing traditional methods. By optimizing several key factors could strengthen the throughput measure to stabilize more data within a timely and efficient manner. On the other hand, the existing HDLRCBSF model shows poor throughput performance, in which it consumes high more training time and raises high communication overhead in large datasets. In this developed model, the throughput analysis is well suited in the financial transaction system to improve the efficiency in larger datasets.

Experimental analysis of the developed model in terms of varied attack scenario

The performance evaluation of the developed model by comparing with existing traditional techniques for various attack scenarios like attack and attack free is shown in Fig. 20. Generally, the attack analysis helps to examine the potential bottlenecks in the system. On the other hand, the validation of attack free helps to analyze whether the system does not contains any malicious activities. This analysis is mainly focused on considering the accuracy analysis to detect the attacks in an effective manner. In blockchain technology, the attack detection helps to provide the significant benefits in order to mitigate the potential damage for managing the integrity of

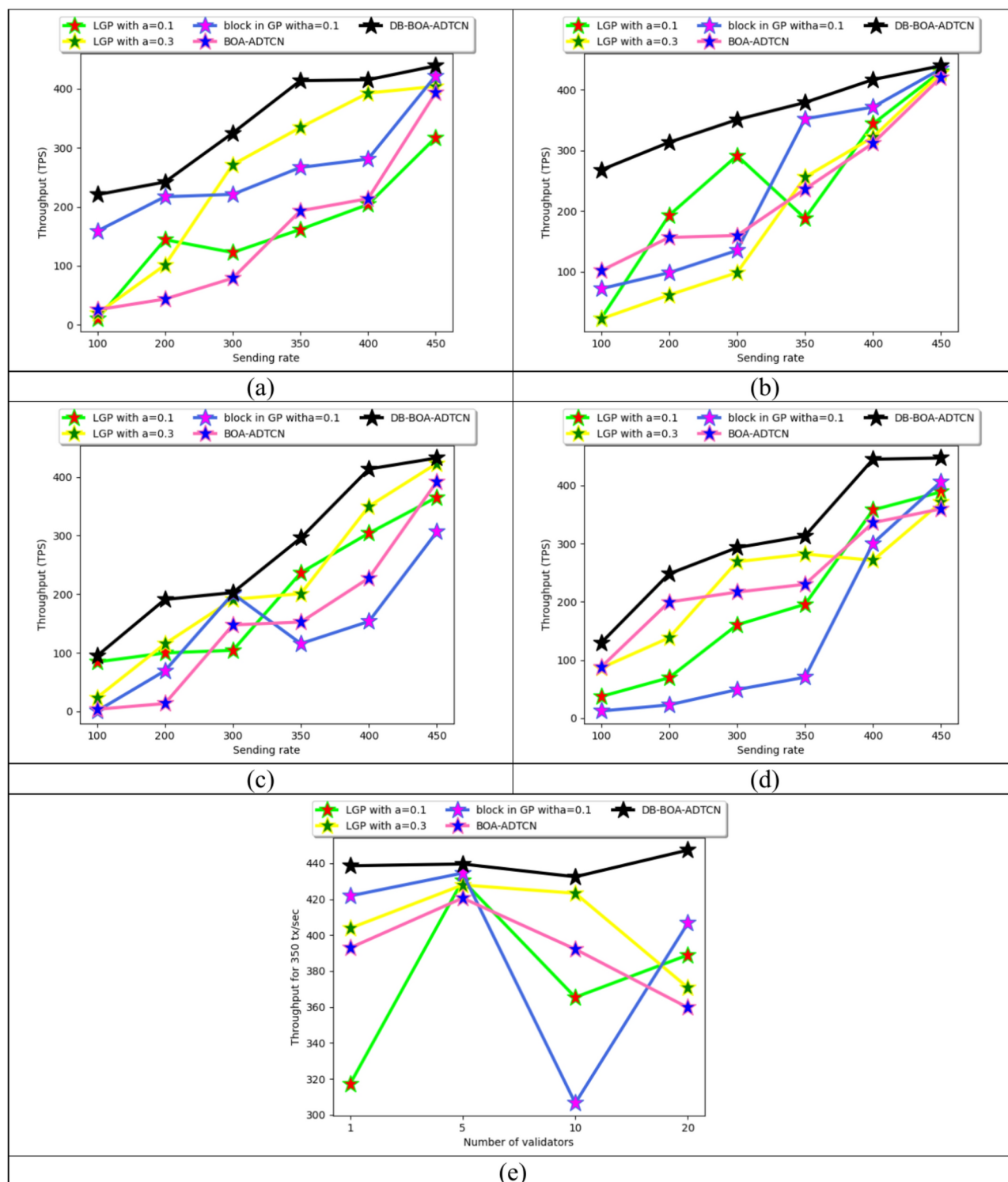


Fig. 14. Throughput analysis of recommended framework in terms of (a) 1 validator (b) 5 validator, (c) 10 validator, (d) 20 validator, (e) 350 validators.

the model to detect the vulnerabilities in an effective manner. However, the large number of data in the existing techniques is ineffective to accurately detect the attacks within the required amount of time.

Conclusion

This work has classified the secure financial data using deep-learning smart contracts in PEC-BC with the proposed DB-BOA. Initially, customer financial data was collected and stored in the BC. Here the leader block

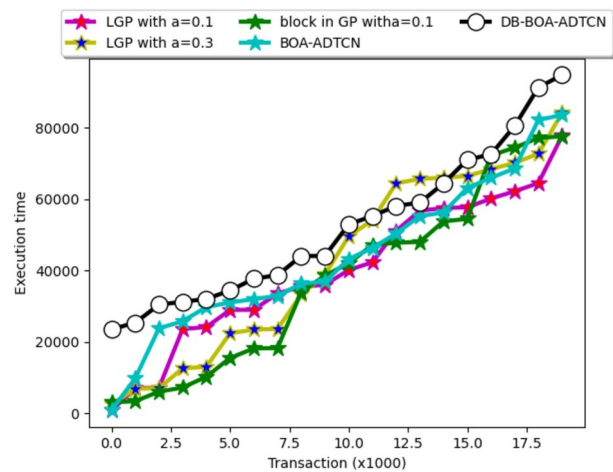


Fig. 15. Execution time analysis for the financial security of the developed model with the other classical algorithm based on transaction.

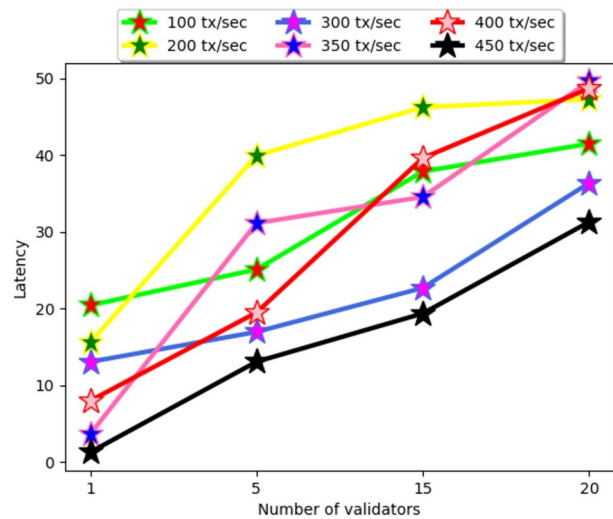


Fig. 16. Execution time analysis for the financial security of the developed model with the other classical algorithm based on transaction.

TERMS	MBO-ADTCN ³⁶	WSA-ADTCN ³⁷	DBOA-ADTCN ²⁹	BOA-ADTCN ³⁰	DB-BOA-ADTCN
Best	1.581792916	1.872703859	1.686221188	1.746616344	1.41555
Std	0.557147214	0.570356531	0.598871372	0.046619772	0.571766
Worst	3.606389111	4.301357614	4.140473104	2.079614712	5.149755
Median	1.768559317	1.987340855	1.711100027	1.746616344	1.427458
Mean	1.989690747	2.184529007	1.927249125	1.753276312	1.575136

Table 2. Statistical evaluation of the recommended technique for financial security.

was optimally selected by the optimized DB-BOA. Then the leader block utilized the ADTCN with a consensus algorithm for a secure smart contract. The proposed DB-BOA optimized the parameters and achieved financial security. From Table 4, the accuracy of the developed model increased by 6.7% of EfficientNet, 3% of Resnet, 5% of DenseNet, and 0.6% of DTCN, respectively. From Fig. 14e throughput value of the implemented model is maximized by 35% of MBO-ADTCN, 41% of WSA-ADTCN, 29.5% of DBOA-ADTCN and 38% of BOA-ADTCN, respectively, by considering the sending rate as 10. The financial security method performance was evaluated with different measures compared with other traditional algorithms and classifiers. Thus the result revealed that the suggested model has a high-security performance than other models.

Terms	MBO-ADTCN ³⁶	WSA-ADTCN ³⁷	DBOA-ADTCN ²⁹	BOA-ADTCN ³⁰	DB-BOA-ADTCN
Accuracy	88.4	91.85	90	93.8	95.45
Sensitivity	88.16986855	92.1132457	89.78766431	94.03437816	95.34884
Specificity	88.62512364	91.59248269	90.20771513	93.57072206	95.54896
Precision	88.3485309	91.46586345	89.96960486	93.46733668	95.44534
FPR	11.37487636	8.40751731	9.792284866	6.429277943	4.451039
FNR	11.83013145	7.886754297	10.21233569	5.96562184	4.651163
NPV	88.62512364	91.59248269	90.20771513	93.57072206	95.54896
FDR	11.6514691	8.534136546	10.03039514	6.532663317	4.554656
F1-score	88.25910931	91.7884131	89.87854251	93.75	95.39707
MCC	0.767968356	0.837013337	0.799972996	0.87600895	0.908988

Table 3. Performance analysis of the developed algorithm.

Terms	Efficientnet ³⁸	Resnet ³⁹	Densenet ⁴⁰	DTCN ⁴¹	DB-BOA-ADTCN
Accuracy	89.05	92.55	90.65	94.8	95.45
Sensitivity	88.97876643	92.61880688	90.59656218	94.7421638	95.34884
Specificity	89.11968348	92.48269041	90.70227498	94.85657765	95.54896
Precision	88.88888889	92.33870968	90.50505051	94.7421638	95.44534
FPR	10.88031652	7.517309594	9.297725025	5.143422354	4.451039
FNR	11.02123357	7.381193124	9.403437816	5.257836198	4.651163
NPV	89.11968348	92.48269041	90.70227498	94.85657765	95.54896
FDR	11.11111111	7.661290323	9.494949495	5.257836198	4.554656
F1-score	88.93380495	92.47854619	90.55078322	94.7421638	95.39707
MCC	0.780976298	0.850990717	0.812979834	0.895987414	0.908988

Table 4. Overall Performance of the developed ADTCN compared over different classifiers.

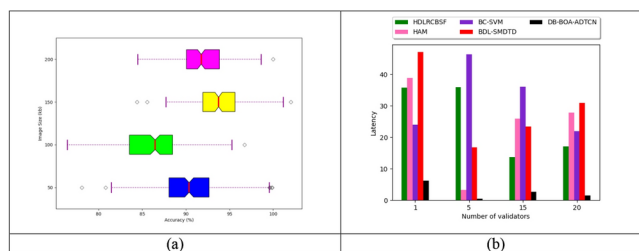


Fig. 17. Scalability analysis of the developed model in financial security system in terms of (a) scalability and (b) latency.

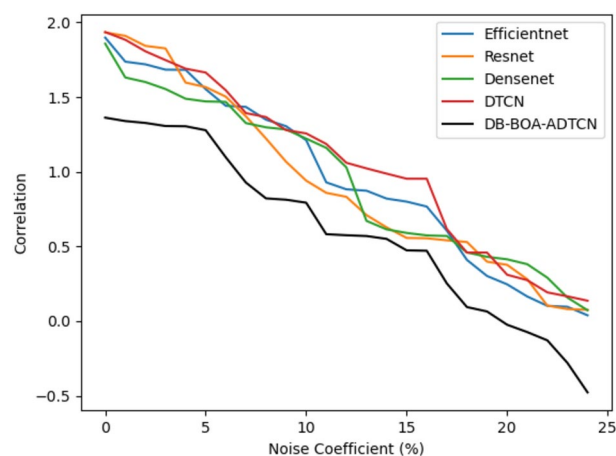
Future work

In order to ensure the non-repudiation and traceability of financial transactions in consortium blockchain, it will be effectively focused and validated in the upcoming works in order to trust the diverse levels and operational cost for different consortium members by adopting the deep learning models. Also, the reduction of communication overhead by comparing with traditional blockchain-based financial systems will be considered and validated in the upcoming works. Optimizing computational resource utilization and the speed of consensus to ensure the feasibility of high volume, mission-critical blockchain-based financial system using the hybrid optimization model will be focused on upcoming works. Focusing Liu et al. approach in the developed model will be further integrated to enhance the scalability, reliability, and efficiency in financial security systems. However, it helps to dynamically adjust the market conditions to maximize the user experience performance in the system.

Algorithm	Computational time (secs)
Algorithms	
MBO ³⁶	15.9653
WSA-ADTCN ³⁷	16.9534
DBOA-ADTCN ²⁹	18.5446
BOA-ADTCN ³⁰	15.6567
Proposed DB-BOA	13.003350
Methods	
EfficientNet ³⁸	14.9848
Resnet ³⁹	16.1698
DenseNet ⁴⁰	18.6417
DTCN ⁴¹	14.3867
Developed ADTCN	13.003350

Table 5. Computational time of the developed model.

Algorithm	Computational complexity
MBO ³⁶	$O[Maxiter + N_{pop} + 2 + chlen]$
WSA ³⁷	$O[Maxiter + 1 + N_{pop} + 1 + chlen]$
DBOA ²⁹	$O[Maxiter + N_{pop} + 2 + chlen + 3]$
BOA ³⁰	$O[Maxiter + N_{pop} + 2 + chlen + 2]$
Proposed DB-BOA	$O[Maxiter + N_{pop} + chlen]$

Table 6. Computational complexity of the developed model in financial security system.**Fig. 18.** Robustness analysis of the developed model of financial security system.

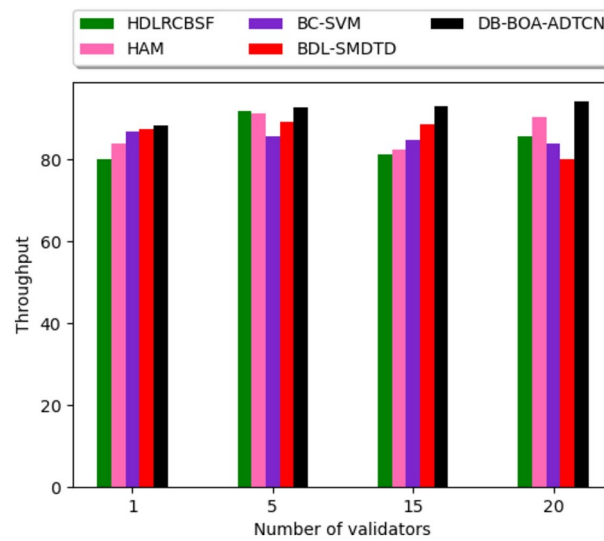


Fig. 19. Throughput analysis of the developed model.

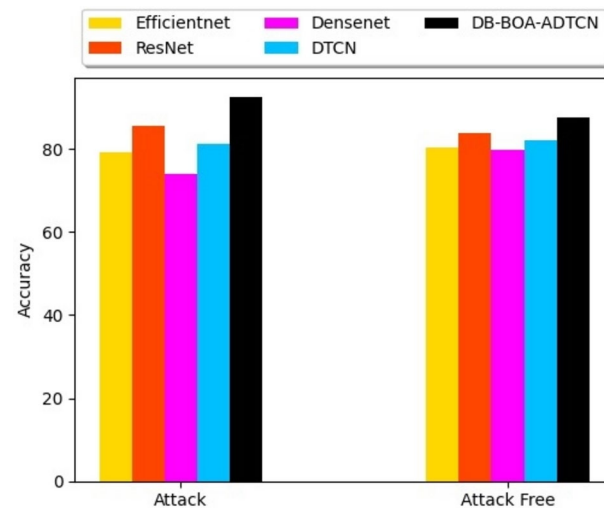


Fig. 20. Performance evaluation of varied attack scenarios of the developed model.

Data availability

The dataset generated and/or analyzed during the current study is not publicly available due to privacy concerns. However, access to the dataset may be granted on a case-by-case basis. Interested researchers can request access by contacting the corresponding author using the contact details provided.

Received: 26 September 2024; Accepted: 5 February 2025

Published online: 25 February 2025

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Acknowledgements

I would like to express my very great appreciation to the co-authors of this manuscript for their valuable and constructive suggestions during the planning and development of this research work.

Author contributions

All authors have made substantial contributions to conception and design, revising the manuscript, and the final approval of the version to be published. Also, all authors agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Funding

This research did not receive any specific funding.

Declarations

Competing interests

The authors declare no competing interests.

Informed consent

Not Applicable.

Additional information

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